A1 | Swin-T on CIFAR-100

Source: A1/run_a1.py + torchvision.models.swin_t • 27,596,254 parameters

RGB 32×32 → resize 96×96
ImageNet normalization

→

Patch embedding: Conv 4×4, stride 4
Permute + LayerNorm → 24 × 24 × 96

HIERARCHICAL STAGES • Shapes: H × W × C • Patch embedding enters Stage 1

Stage 1: 2 blocks
3 heads
24 × 24 × 96

→

Stage 2: 2 blocks
6 heads
12 × 12 × 192

→

Stage 3: 6 blocks
12 heads
6 × 6 × 384

→

Stage 4: 2 blocks
24 heads
3 × 3 × 768

Between stages: Patch Merging = concatenate 2×2 tokens → LayerNorm(4C) → Linear 4C → 2C.

LayerNorm → permute to C × H × W → global average pool → 768 values
Linear head 768 → 100 → 100 logits

SWIN BLOCK • Pre-norm residual paths

$$Y = X + \mathrm{StochasticDepth}(\mathrm{WindowAttention}(\mathrm{LayerNorm}(X)))$$
$$Z = Y + \mathrm{StochasticDepth}(\mathrm{MLP}(\mathrm{LayerNorm}(Y)))$$

• MLP: C → 4C → GELU → C

Attention uses 7×7 windows with relative position bias; successive blocks use shifts 0 and 3.
Window padding: 24→28, 12→14, 6→7 and 3→7; outputs are cropped back after attention.
At 96×96 input, shifts are disabled in the 6×6 and 3×3 stages (one padded window).
Patch merging is separate from attention padding. There is no CLS token.
Partial fine-tuning trains features[-1] (Stage 4) + head; the final norm stays frozen.